

# CS 486/686

## Dissecting a Small Language Model

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Lecture 21

Opening the hood on Qwen3-0.6B

Search ›

Uncertainty ›

Decisions ›

Learning

## Learning goals

- Load or inspect a **small causal LM**.
- Trace one generated token at a time.
- See how **decoding** and **prompting** change behavior.
- Recognize limitations and failure modes.

## From L20 to L21



L20 taught the pipeline. Today we watch it run.

# The model we will dissect

**Qwen3-0.6B**

a real small causal language model

**0.6B**

parameters

**28**

layers

**32k**

context

**~151k**

vocab

**GQA**

16 Q / 8 KV

**BBPE**

tokenizer

A "small" LM is still a serious learned system.

# What gets loaded?

## tokenizer

text ↔  
token IDs

## config

layers,  
heads, vocab

## weights

learned  
parameters

## settings

temperature,  
max tokens

# Demo path and fallback

## optional live path

Transformers.js +  
ONNX + WebGPU

## reliable path

precomputed traces  
embedded here

The concepts do not depend on a live GPU demo working.

# Optional browser inference

```
import { pipeline, TextStreamer } from "@huggingface/transformers";  
  
const generator = await pipeline(  
  "text-generation",  
  "onnx-community/Qwen3-0.6B-ONNX",  
  { device: "webgpu", dtype: "q4f16" },  
);
```

This loads a web-ready tokenizer and quantized model weights.

# The same pipeline in Python

```
tok = AutoTokenizer.from_pretrained("Qwen/Qwen3-0.6B")
model = AutoModelForCausalLM.from_pretrained(
    "Qwen/Qwen3-0.6B",
    torch_dtype="auto",
    device_map="auto",
)
```

Browser and Python paths expose the same conceptual pieces.

## Start with a prompt

Explain gradient descent in one sentence.

Inference begins as human-readable text.

# The model sees a chat template

system

You are a helpful assistant.

user

Explain gradient descent in one sentence.

assista  
nt

The model's actual input is usually not exactly what the user typed.

# What the model actually receives LIVE

The chat template wraps your text in roles and special tokens.

What the user types vs. what the model actually receives: roles and special tokens added by the chat template.

"Explain gradient descent in one sentence."

"Who won the 2031 Turing Award?"

thinking: OFF

```
<|im_start|>user
Explain gradient descent in one sentence.<|im_end|>
<|im_start|>assistant
<think>

</think>
```

Thinking mode adds a reasoning section the model fills in before its answer.

# Prompt to tokens

Explain gradient descent in one sentence.

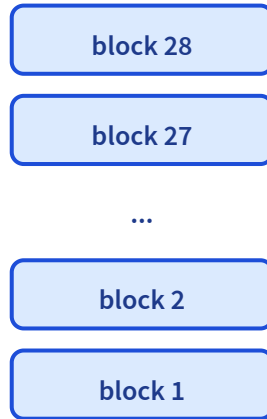
| token    | illustrative id |
|----------|-----------------|
| Explain  | 8493            |
| gradient | 33497           |
| descent  | 41184           |

# Tokens become embeddings



Each token begins generation as a learned vector.

# Through 28 transformer blocks



**Attention:** communicate across context.

**Feed-forward:** per-token computation.

Residuals and normalization  
keep the deep stack stable.

# Inside the blocks: real attention

LIVE MODEL

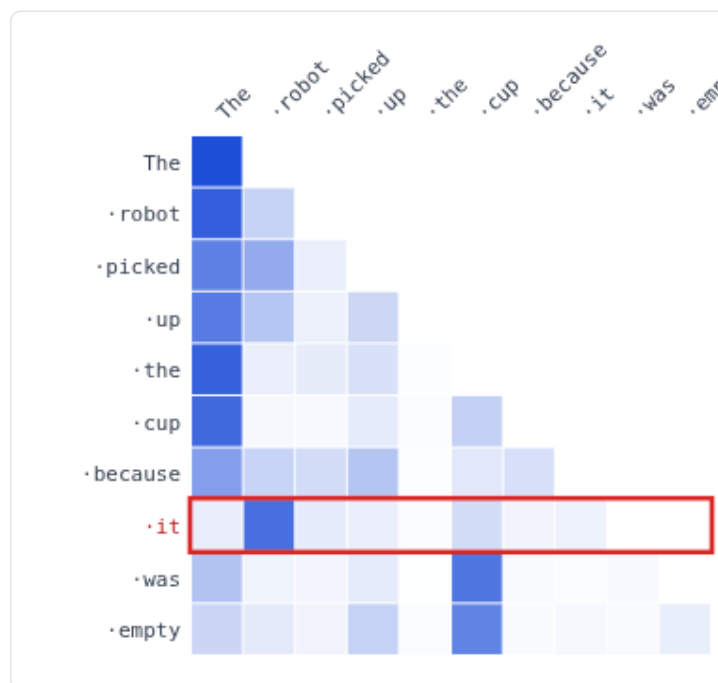
Actual Qwen3-0.6B attention — pick a head, click a query token.

Real attention from inside **Qwen3-0.6B** (28 layers, 16 heads). Different layer/head pairs specialize.

L11 H1: it → robot

L15 H3: previous token

L0 H6: broad context



query: **it** attends to...

robot 0.74

cup 0.10

picked 0.04

The 0.03

up 0.03

it 0.03

## The last position predicts next



For generation, the next distribution comes from the final prompt position.

# Logits: top candidates

Illustrative trace after the prompt: Explain  
gradient descent in one sentence.

| candidate token | logit | interpretation |
|-----------------|-------|----------------|
| Gradient        | 8.4   | high           |
| It              | 6.8   | medium         |
| A               | 6.3   | medium         |
| zq              | -2.5  | very low       |

The exact values here are a teaching trace, not a live benchmark.

# Softmax and decoding choice

Gradient

0.54  
It

0.21  
A

0.16  
zq

# Watch it run

LIVE MODEL

A real Qwen3-0.6B answer (precomputed); temperature and thinking are yours to change, then run it live.

"Explain gradient descent in one sentence."

"Who won the 2031 Turing Award?"

temperature  0.7

thinking: OFF

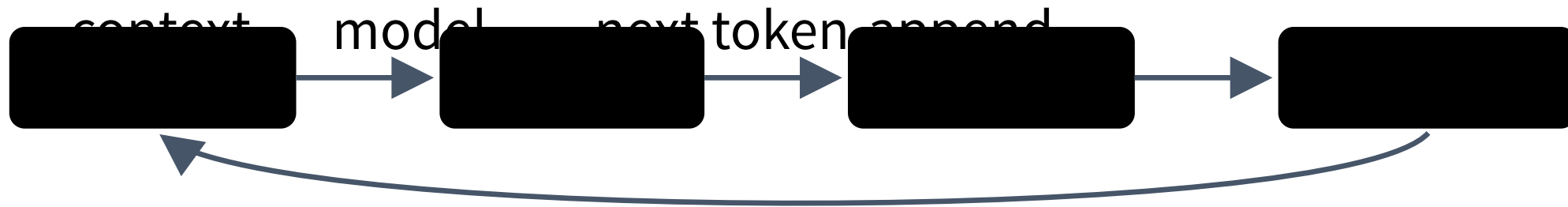
run (real model)

prompt: Explain gradient descent in one sentence.

|          |                                                                                      |      |
|----------|--------------------------------------------------------------------------------------|------|
| Gradient |    | 0.99 |
| The      |    | 0.00 |
| A        |    | 0.00 |
| Gr       |   | 0.00 |
| Grad     |  | 0.00 |
| In       |  | 0.00 |

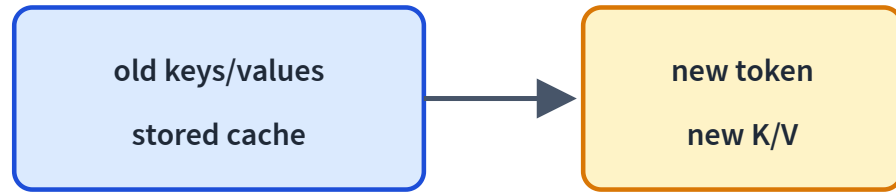
Gradient descent  
is a method used  
to minimize a  
function by  
iteratively  
adjusting the  
parameters of a  
model in the

# Append and repeat



Generation is autoregressive and sequential at inference time.

# KV cache: do not recompute everything

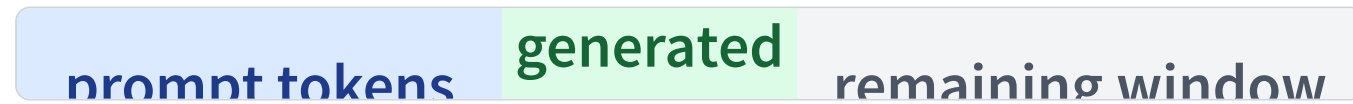


Generated tokens  
keep their keys/values.

The next step computes  
only the new token's pieces.

This is why long  
context costs memory.

# Context length is working memory



For Qwen3-0.6B: up to about **32k tokens**.

It is not permanent memory; it is the current working context.

# Decoding changes behavior

## greedy

deterministic, safe, repetitive

## sampling

varied, sometimes  
better, sometimes worse

Weights stay fixed; only the token selection rule changes.

# Temperature controls randomness

**low T**

Gradient



It



A



**high T**

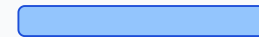
Gradient



It



A



# Top-k and top-p trim the tail

## top-k

sample from the  $k$   
most likely tokens

## top-p

sample from tokens whose  
cumulative mass reaches  $p$

Keep randomness inside plausible choices.

# Prompting is inference-time conditioning

## Prompt A

Explain gradient descent.

## Prompt B

Explain gradient descent to a 10-year-old.

The context changes. The weights do not.

# Few-shot prompting specifies a pattern

```
tiny → small  
huge → big  
rapid → fast
```

Examples in the prompt can define a task without training.

# Chain-of-thought-style prompting

## direct

Answer the question.

## reasoning style

Think step by step, then answer.

Reasoning traces can help behavior, but are not guaranteed faithful explanations of internal computation.

## Some models expose thinking modes

**non-thinking**

answer directly

**thinking**

produce intermediate  
reasoning text

For Qwen-style models, special chat-template controls can change this behavior.

## Failure mode: hallucination

Who won the 2031 Turing Award?

A model may answer fluently anyway.

Next-token likelihood is not truth verification.

# Failure mode: prompt sensitivity

## Prompt 1

Give a concise answer.

## Prompt 2

Be creative and speculate.

The model is conditioned by text; small changes can matter.

# Failure mode: context and cost

## context

long prompts  
consume the window

## latency

generation is  
token-by-token

## browser

hardware  
support varies

# What did not change today?

**no fine-tuning**

weights unchanged

**no LoRA**

no adapters

**no retrieval**

no external documents

We steered behavior only through inputs and decoding.

## Next question

When is prompting not enough, and when do we update weights or add external knowledge?

Can another model **build those new weights** for us?

L22: this same Qwen3-0.6B becomes a frozen interpreter for compiled functions.

# Recap

- ✓ Loaded tokenizer, config, weights, and generation settings.
- ✓ Tokenized a prompt and traced it into embeddings.
- ✓ Ran a causal LM to produce logits.
- ✓ Decoded one token at a time.
- ✓ Used prompting and decoding without updating weights.
- ✓ Identified common failure modes.